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Explaining Condition-Based Maintenance for Metro Trains Using Ensemble ML with XAI and Hybrid Deep Learning Models

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***Abstract*— Metro train systems serve as a backbone for urban transportation but are increasingly vulnerable to operational failures, which can disrupt services and compromise passenger safety. Ensuring their reliability requires robust and accurate methods for failure classification and remaining useful life (RUL) estimation. However, existing approaches often face significant challenges, including imbalanced datasets, high computational costs, reliance on synthetic data, difficulties in processing large time-series datasets, and risks of overfitting.**

To address these challenges, this research proposes a comprehensive framework of ensemble machine learning methods like Stacking, Random Forest, Bagging, and XGBoost and advanced hybrid deep learning models like TCN-GRU which combines the temporal feature extraction capability of Temporal Convolutional Networks (TCNs) with the sequential modelling efficiency of Gated Recurrent Units (GRUs). Furthermore, resampling techniques like down-sampling for large time series dataset were employed to retain over 90% of the original data variance, significantly reducing computational overhead. Explainable AI (XAI) tools, such as SHAP and LIME, were integrated only into the ensemble methods to enhance interpretability.

The proposed framework was evaluated on a real-world metro train dataset, with a focus on the air production unit due to its critical role in train brakes. Results demonstrated that ensemble methods like Stacking performed robustly in failure classification, and the hybrid TCN-GRU model achieved the best results in remaining useful life (RUL) prediction. Also, the results shows that the integration of SHAP offers actionable insights into ensemble models like xgboost, allowing maintenance teams to identify and adjust key variables, thus reducing the chances of future failures.

This study underscores the potential of ensemble and hybrid models for advancing predictive maintenance in public transportation. The proposed approach offers significant benefits, including cost reductions, enhanced reliability, optimized performance, and improved safety. Future research will focus on extending these methods to other industrial systems, emphasizing real-time deployment and adaptive learning for broader applicability.

Keywords—Predictive maintenance, Condition-based maintenance, metro trains, ensemble machine learning, hybrid deep learning, TCN-GRU model, Explainable AI, failure classification, remaining useful life estimation.

I. INTRODUCTION

In Industrial systems such as metro trains [1][2][3], elevator systems [6], rotating machines [7], trucks [11], etc., there exists a possibility for future failures that may endanger the safety of users and operators. To tackle these issues, the idea of predictive maintenance has been extensively utilized to predict and classify various kinds of failures and to assess their remaining useful life (RUL) [4][15]. This is accomplished by implementing traditional machine learning and deep learning algorithms across different machines. This paper takes a similar approach, using condition-based maintenance to evaluate historical data and identify failure patterns specifically for metro trains. Ensuring the reliability and efficiency of metro trains is critical to minimize service disruptions, optimize machine performance, enhance reliability, lower maintenance costs, reduce unexpected breakdowns, and improve passenger and operator safety in public transportation systems.

Despite advancements in predictive maintenance research, current approaches encounter multiple limitations. These include difficulties in managing imbalanced datasets [11][12], handling large multivariate time-series data [1][5], incorporating explainable AI (XAI) algorithms [7][15], dependence on synthetic data [12][11], restricted access to labeled data [1][13], a limited focus on single machine components [4][5][13], high computational costs related to large datasets [9][10], and constraints in real-time implementation [1][2][4][13]. These challenges underscore the necessity for innovative strategies in condition-based maintenance for industrial machinery.

Our main objective is to resolve these difficulties by utilizing ensemble [9][12][15], hybrid [5][9][13], and XAI [7][10][15] models. First, a majority of studies have employed techniques like SMOTE [11][12], oversampling, and undersampling [11] to address class imbalance in datasets. However, these approaches frequently produce synthetic data, which may not be dependable for predicting failures, especially in critical scenarios such as metro trains. Instead, this research utilizes high-performing ensemble machine learning models like Random Forest, Gradient Boosting, Bagging, and their stacking variations. These models attribute weights to minority classes, allowing for greater emphasis on these classes during predictions.

Second, by implementing advanced deep learning architectures such as CNN, LSTM, GRU, and TCN, we proposed hybrid models like TCN-GRU. This integration capitalizes on the advantages of TCN in extracting temporal

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features and GRU in efficiently modeling sequential dependencies within time-series datasets. These models improve prediction accuracy while reducing training duration and decreasing the chances of overfitting and underfitting.

To manage large time-series datasets and lower computational costs, we adopt resampling techniques that decrease dataset size while preserving approximately 90% of the variance from the original data. This approach ensures that prediction accuracy remains unaffected. Furthermore, unlike numerous studies that depend on synthetic datasets generated by generative models, we utilize real-world data from the UCI repository, which features labeled datasets obtained from metro train components. We preprocess this data using methods like z-score normalization to guarantee its quality and reliability. There will always be obstacles in real-world data collection in this area, as it requires numerous sensors, machines, time, patience, etc.

Finally, we incorporate XAI techniques to globally identify and visualize the most significant parameters in failure predictions, both for failure classification and RUL estimation. This improves interpretability and allows train maintenance staff to make well-informed adjustments, thereby decreasing the probability of future failures.

In summary, if we clearly state the **problem statement** of our study, we aim to address the shortcomings of current research by creating a precise and interpretable condition-based maintenance system for metro trains. We utilize ensemble machine learning and hybrid deep learning algorithms on authentic historical time-series data to enhance failure classification and RUL estimation.

The core research questions and contributions addressed in this study are as follows:

- How can ensemble machine learning models (e.g., Random Forest, Gradient Boosting, Bagging) be effectively utilized to address class imbalance and other limitations in metro train failure prediction without relying on synthetic data generation?
- What are the most effective hybrid deep learning models (e.g., TCN-GRU) for improving prediction accuracy while minimizing training time and avoiding overfitting or underfitting?
- How can time-series data resampling techniques be applied to large datasets in metro train failure prediction to reduce computational costs while preserving high variance and accuracy?
- How can XAI methods be integrated into failure prediction models to identify and visualize the most critical variables, improving interpretability and aiding maintenance staff in decision-making?

II. Literature Review

Davari et al. [1] focus on detecting failures in air production units in the railway industry to prevent breakdowns. They propose a data-driven anomaly detection framework using a sparse autoencoder and a low-pass filter for failure detection. Challenges include limited labeled data, real-time implementation, and handling multivariate time-series data. Future directions suggest improving labeled datasets, optimizing real-time implementation, and advancing data handling methods.

Neamah and Bayir [2] explore fault prediction in industrial systems like MetroPT using machine learning (ML) algorithms, including SVM, KNN, and reinforcement learning. Their framework achieves high accuracy, but scalability and real-world validation remain unaddressed. Future work involves testing on larger datasets, integrating real-time data, and developing scalable models.

Shinde et al. [3] address predictive maintenance in rail transportation using ML and DL methods like LSTM, Random Forest, and Variational Autoencoders. They highlight limitations in system completeness, precision-recall trade-offs, and sample size. Future research could apply metaheuristic methods, consider risk factors, and test real-world implementations.

Hermawan et al. [4] predict the remaining useful life (RUL) of aircraft engines using a CNN-LSTM hybrid model to analyze historical data. Challenges include reliance on single-engine datasets and limited real-world validation. Future research should explore diverse datasets and energy-efficient approaches for real-time applications.

Du et al. [5] propose a Transferable CNN-LSTM model for predicting mechanical equipment failures. The study employs Domain-Adversarial Neural Networks (DANN) for model portability but lacks real-world validation and data quality considerations. Future directions include integrating domain knowledge, enhancing preprocessing techniques, and adopting adaptive learning methods.

Verma et al. [6] predict faults in elevator systems using deep learning methods like LSTM and Random Forest with sensor data. Limitations include uncertainties in real-world conditions and neglecting external factors like holidays. Future research could address model uncertainties and adapt to dynamic operational conditions.

Gawde et al. [7] utilize Explainable AI (X-AI) models, such as LIME and SHAP, for predicting failures in industrial rotating machines. The study emphasizes interpreting multi-sensor data but lacks combinations of X-AI models and fails to address multiple fault types. Future work involves improving X-AI methods and exploring multi-fault prediction models.

Pradeep et al. [8] focus on semiconductor manufacturing failures using ML and neural network models on the Secom dataset. Key challenges include resource shortages and practical implementation constraints. Future directions suggest analyzing resource limitations and applying hybrid methods for better prediction accuracy.

Al-Andoli et al. [9] develop a parallel ensemble model combining ML and DL techniques to improve industrial fault prediction. While achieving high accuracy, challenges include high computational costs from gradient descent methods. Future work may explore efficient optimization techniques and broader algorithm applications.

Pham et al. [10] classify failures in ball-bearing systems using ensemble methods like Random Forest and Gradient Boosting. The study employs SHAP for interpretability but suffers from computational inefficiencies and lacks adaptability. Future research could develop flexible, adaptive algorithms and refine interpretable models for high-dimensional data.

Selvi et al. [11] investigate air brake system failures in Scania trucks, addressing class imbalance with resampling techniques like SMOTE. Classification methods such as Logistic Regression and SVM are used, but attribute selection methods are not optimized. Future work should focus on advanced attribute selection and balancing techniques.

Elahi et al. [12] improve software failure predictions using ensemble methods like model averaging and stacking. They employ resampling techniques like SMOTE but rely on datasets from the PROMISE repository, limiting generalizability. Future research could use real-world datasets and develop better classifier combinations for fault prediction.

Bundasak and Wittayasirikul [13] predict motor health using algorithms like Logistic Regression, KNN, and SVM. The study integrates a web-based notification system but faces challenges in collecting large datasets and handling noise. Future work could streamline data collection and improve scalability for larger datasets.

Mofidul et al. [14] predict failures in solar energy systems using ensemble models like XGBoost and LightGBM. They achieve high accuracy but lack comprehensive feature exploration. Future directions include integrating additional features and refining ensemble techniques for predictive modeling.

Khan et al. [15] propose an X-AI-based regression framework using LIME and SHAP to predict the RUL of industrial machinery. Challenges include balancing accuracy with interpretability and limited exploration of other X-AI techniques. Future research should expand to diverse datasets, explore alternative X-AI methods, and enhance model clarity.

III. METHODOLOGY

The study does the classification of failure and predict remaining useful life using ensemble machine learning models and hybrid deep learning models for the MetroPT-3 Dataset [16]. The methods or steps used to implement this is given in below Figure 1.

A. Data Collection

The MetroPT-3 dataset [16], sourced from the UCI repository, originates from a study [1] in Porto, Portugal, focusing on the Air Production Unit (APU) of metro trains. It consists of 1.5 million multivariate time-series data points recorded from February to September 2020, with a sampling rate of 0.1 Hz meaning each reading was taken after 10 seconds. The dataset includes a timestamp column and readings from 7 analog sensors and 8 digital sensors that monitor 15 key signals, such as timestamp, TP2, TP3, H1, DV_pressure, Reservoirs, Oil_temperature, Motor_current, COMP, DV_eletric, Towers, MPG, LPS, Pressure_switch, Oil_level and Caudal_impulses. The APU comprises components like the Air Intake Valve, Air Filter, Air Compressor, Air Tanks, and Air Drying System, which are critical for train operations like braking and door control.

7 Analog sensors measure continuous variables such as pressure at various points (compressor, pneumatic panel, filters, reservoirs), motor current, and oil temperature. 8 Digital sensors provide binary on/off signals for events like compressor activity, air discharge, drying tower operation, pressure drops, and oil levels. A detailed description for each of the 15 parameters and APU components is available in the paper [1].

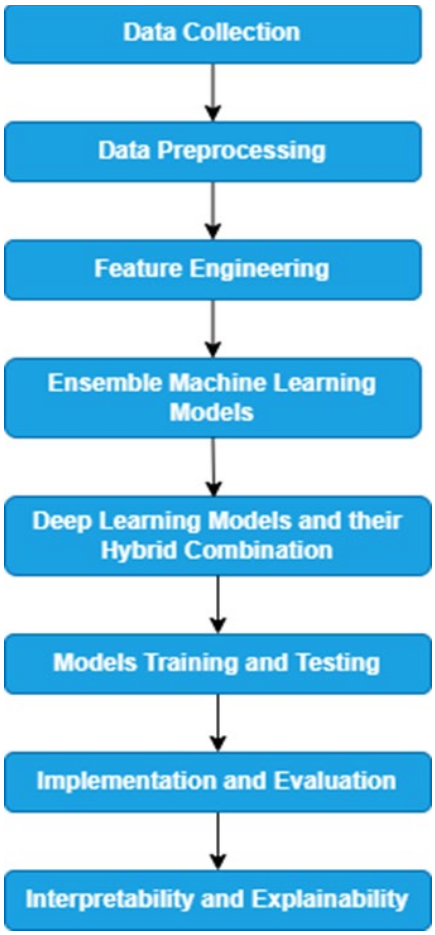


Fig. 1. Methodology

For labelling the "Failure" column for failure classification and labelling "Hours till Failure" column for RUL prediction in above dataset, we have used the failure report provided by the same paper [1] and from MetroPT-3 dataset UCI repository website. How authors form paper [1] has generated this failure report is detailed explained in the same paper. Below Figure 2 shows the failure report from paper [1] in detail:

Nr.	Start Time	End Time	Failure	Severity
#1	4/18/2020 0:00	4/18/2020 23:59	Air leak	High stress
#1	5/29/2020 23:30	5/30/2020 6:00	Air Leak	High stress
#3	6/5/2020 10:00	6/7/2020 14:30	Air Leak	High stress
#4	7/15/2020 14:30	7/15/2020 19:00	Air Leak	High stress

Fig. 2. Failure Report available in Paper [1].

Firstly, timestamp column of dataset and start and end timestamp in failure report was converted to datetime format. Then, the "Failure" column in the df dataset is labeled by comparing each timestamp against predefined failure periods from the failure report dataFrame. If a timestamp falls within a specific failure period range provided in failure report, the corresponding failure type "AirLeak" is assigned based on the report. Otherwise, it defaults to "NoAirLeak". We can use same logic for labelling "Severity" column into high, low and medium stress.

The "Hours_till_Failure" column is labelled and calculated by determining the time difference between each timestamp in the dataset and the start time of the nearest upcoming failure period from the failure report, provided the

timestamp is earlier than the failure period's start time. This time difference is then converted into hours. If the timestamp falls within an active failure period, the value is set to 0. For timestamps after all failure periods, the value defaults to 0, as failures are considered recorded. This logic is implemented in a function that was applied to the timestamps in the df dataset, creating the "Hours_till_Failure" column.

The resulting dataset after labelling will look like below Figure 3, some columns we have not mentioned in Figure 3 as it takes up huge space having overall 19 columns.

	timestamp	TP2	TP3	H1	...	Failure	Hours_till_Failure
0	2020-02-01 00:00:00	-0.012	9.358	9.340	...	NoAirLeak	1848.000000
1	2020-02-01 00:00:10	-0.014	9.348	9.332	...	NoAirLeak	1847.997222
2	2020-02-01 00:00:19	-0.012	9.338	9.322	...	NoAirLeak	1847.994722
3	2020-02-01 00:00:29	-0.012	9.328	9.312	...	NoAirLeak	1847.991944
4	2020-02-01 00:00:39	-0.012	9.318	9.302	...	NoAirLeak	1847.989167
5	2020-02-01 00:00:49	-0.012	9.306	9.290	...	NoAirLeak	1847.986389
6	2020-02-01 00:00:59	-0.012	9.296	9.280	...	NoAirLeak	1847.983611
7	2020-02-01 00:01:09	-0.014	9.286	9.270	...	NoAirLeak	1847.980833
8	2020-02-01 00:01:19	-0.012	9.276	9.258	...	NoAirLeak	1847.978056
9	2020-02-01 00:01:29	-0.012	9.264	9.248	...	NoAirLeak	1847.975278

Fig. 3. MetroPT-3 dataset [16] after labelling.

B. Data Preprocessing

The initial dataset was free of missing values. Preprocessing steps were performed as follows:

1. Data Encoding: After labelling the dataset, we applied manual mapping for data encoding in the "Failure" column, assigning "AirLeak" a value of 1 and "NoAirLeak" a value of 0 to facilitate accurate classification of failures in different models. The same method was used for the "Severity" column, mapping "Low" to 0, "Medium" to 1, and "High" to 2.

2. Time series dataset resampling (down-sampling): After encoding the dataset, we addressed issues such as converting improper timestamp data into a proper sequence and reducing the size of the dataset, which initially contained around 1.5 million records. Given that some sensor readings in each column were similar, we performed 2-minute (2T) downsampling using the resample function. This reduced the dataset from 1,516,948 to 126,527 records. During this process, we aggregated the numerical sensor columns, such as TP2, TP3, H1, Oil_temperature, etc. by calculating their mean values. For digital sensor columns, such as COMP, DV_electric, etc. we used the mode value due to their binary nature (0 and 1). After resampling, some of the entire records has NaN (missing) values appeared in the whole dataset, which were subsequently dropped. These missing values resulted from improperly formatted timestamp data converted into proper format during the resampling process, which caused missing records in the dataset. The variance percentage for each column after resampling, compared to the original dataset calculated, ranges from about 3.9% (for Pressure_switch) to 103.1% (for MPG), with most columns retaining over 90% of their variance. This indicates that the resampling process did not significantly affect the variability of the dataset, which suggests that the resampled dataset should still be suitable for analysis and prediction tasks.

3. Z- score Normalization (standard scaler): Each column's data points were converted into their corresponding Z-score values using the Z-score formula. This process results in a mean of 0 and a standard deviation of 1 for each column. While some classical models require scaling before making predictions to ensure the data is centered around 0 and has a consistent scale, this is not necessary for failure classification tasks. This is because we used an ensemble and hybrid method, which can handle data with inconsistent scales. However, Z-score normalization was applied specifically for predicting "Hours_till_Failure" due to the range of values in this feature, which spans from 0 to 1847.98. Without scaling, this wide range could lead to high errors in the final remaining useful life predictions. After scaling and making predictions, if we wish to view the original predicted “Hours_till_Failure” values, we can use the inverse transform function available in Python's standard scaler.

C. Feature Engineering

After data preprocessing, correlation heatmap was used to find the correlation between each of the columns with the target column in pre-processed dataset. Based on this analysis, pre-processed dataset having these columns like timestamp, Severity, Failure, and Hours_till_Failure was dropped and remaining features was used for classifying target column “Failure”. Similarly, pre-processed dataset having timestamp column was dropped and rest of the columns was used to predict “Hours_till_Failure” using various models. At the end of this study, we will also use XAI method such as LIME and SHAP to extract most important features which is responsible for final failure classification and remaining useful life predictions.

D. Ensemble Machine Learning Models

To classify failure and predict remaining useful life (Hours_till_Failure), we employed four ensemble models: Random Forest, Extreme Gradient Boosting (XGBoost), Bagging, and Stacking.

1. Random Forest: This model constructs multiple decision trees using random subsets of the data and features. The outputs of these trees are combined through voting (for classification) or averaging (for regression), resulting in improved accuracy and reduced overfitting.

2. Extreme Gradient Boosting (XGBoost): XGBoost combines weak learners (typically decision trees) sequentially. Each subsequent model reduces the errors of its predecessor by optimizing a loss function using gradient descent. Regularization techniques (L1/L2) are applied to prevent overfitting, making it highly effective for complex datasets.

3. Bagging: Bagging trains multiple models on random subsets of the data and aggregates their predictions using voting (for classification) or averaging (for regression). While its architecture is similar to that of Random Forest, Bagging can use any classical model as a base estimator. In this study, we utilized decision trees as the base estimator.

4. Stacking: Stacking combines predictions from multiple base models by employing a meta-model that

learns from their outputs. This approach enhances accuracy by leveraging the strengths of individual models. In our study, Random Forest and XGBoost were used as base models, while XGBoost served as the meta-model.

E. Deep Learning Models and their Hybrid Combination

To classify failures and predict remaining useful life (Hours_till_Failure), we utilized four deep learning models: CNN, LSTM, GRU, and TCN. The best-performing models were combined (TCN-GRU) to create a hybrid model that achieved higher accuracy and reduced training time. Time-series models (LSTM, GRU, and TCN) segmented the dataset into fixed-length sequences (timesteps=10) for analysis. All models employed two dense layers: the first with 64 neurons and ReLU activation, and the second with 1 neuron (sigmoid for classification, none for regression). A dropout rate of 0.2 was applied to dense layers (and between layers in RNN-based models) to mitigate overfitting. Models were optimized for time-series data with appropriate activation functions (ReLU and sigmoid), and pooling layers were avoided in CNN and TCN to retain critical information.

1. CNN (Convolutional Neural Network): CNNs, typically used for image data, were adapted for time-series tasks to extract spatial patterns. Two convolutional layers with 64 filters, a kernel size of 2, ReLU activation, and same padding were used, omitting pooling layers. The output was flattened into 1D before being passed to the dense layers.
2. LSTM (Long Short-Term Memory): LSTMs, a type of RNN, were used to model long-term dependencies in time-series data. They employed cell states and gating mechanisms (input, forget, and output gates). The architecture consisted of two LSTM layers with 64 neurons and ReLU activation, with dropout applied between layers before connecting to the dense layers.
3. GRU (Gated Recurrent Unit): GRUs, a simpler RNN variant, efficiently captured long-term dependencies using gating mechanisms. The model included two GRU layers with 64 neurons and ReLU activation, with dropout applied between layers before feeding into the dense layers.
4. TCN (Temporal Convolutional Network): TCNs leveraged dilated convolutions to capture long-range dependencies, making them ideal for time-series forecasting. Two convolutional layers with 64 filters, a kernel size of 2, ReLU activation, and causal padding were used. Dilation rates were 1 and 2 for the two layers, respectively. The outputs were flattened into 1D before being passed to the dense layers.
5. Hybrid Model (TCN-GRU): The TCN-GRU hybrid combined the strengths of TCNs for extracting long-range temporal features and GRUs for sequential pattern recognition. Outputs from two TCN convolutional layers were fed into two GRU layers, followed by flattening and dense layers. The same parameters and structures as individual TCN and GRU models were used.

F. Models Training and Testing

The pre-processed dataset was split into 80% training data and 20% testing data. Hyperparameter tuning was performed manually rather than using automated algorithms. For ensemble models, parameters such as n_estimators were adjusted for optimal predictions. For deep learning and hybrid models, the Adam optimizer with a learning rate of 0.001 was used during model compilation.

G. Implementation and Evaluation

To identify the best-performing models among ensemble machine learning, deep learning, and hybrid models, various performance metrics were used for both classification and regression tasks. For classification, metrics included accuracy, F1-score, recall, and precision. For regression, metrics included mean squared error (MSE), root mean squared error (RMSE), mean absolute error (MAE), and R2-score.

H. Interpretability and Explainability

For only interpreting decisions of various ensemble machine learning models (e.g. XGBoost model), feature extraction for future model improvements, and identifying key features that maintenance staff can adjust to reduce future train failures, we utilized XAI (Explainable Artificial Intelligence) models, specifically LIME and SHAP. Both methods are model-agnostic and applicable to any model type.

LIME (Local Interpretable Model-Agnostic Explanations): LIME explains individual predictions by approximating complex models with simple, interpretable models (e.g., ensemble models). It perturbs the input data, observes prediction changes, and builds a locally faithful surrogate model for specific predictions.

SHAP (SHapley Additive exPlanations): SHAP uses cooperative game theory to assign each feature an importance value. It calculates Shapley values to represent the average contribution of each feature to the model's prediction, providing consistent and globally accurate explanations for both individual and global model behavior.

IV. Result and Discussion

A. Various Models Performance

The performance of ensemble machine learning models, deep learning models, and a hybrid model for metro train failure classification was evaluated using accuracy, F1-score, precision, and recall metrics, as summarized in Table 1.

TABLE I. Failure Classification Models

Models	Accuracy	F1-score	Precision	Recall
Random Forest	99.59%	99.60%	99.63%	99.59%
XGBoost	99.60%	99.62%	99.66%	99.60%
Bagging	99.59%	99.61%	99.65%	99.59%
Stacking	99.95%	99.96%	99.96%	99.96%
CNN	99.85%	99.85%	99.85%	99.85%
LSTM	99.80%	99.81%	99.82%	99.80%
GRU	99.93%	99.93%	99.93%	99.93%
TCN	99.88%	99.89%	99.89%	99.89%
TCN-GRU	99.90%	99.90%	99.90%	99.90%

Our classifier models demonstrated performance equal to or slightly better than existing approaches. The classifiers models addressed key research objectives and questions by assigning weights to minority classes during training, enabling accurate failure classification without generating synthetic data. Among classifier models, Stacking achieved the highest performance with 99.95% accuracy, precision,

recall, and F1-score, outperforming Random Forest, XGBoost, and Bagging, which all exceeded 99.5%.

In deep learning, GRU delivered the best results (99.93%), followed closely by TCN (99.88%) and CNN (99.85%), while LSTM achieved 99.80%. The hybrid TCN-GRU model slightly improved upon individual deep-learning models, reaching 99.90% across all metrics. This highlights the effectiveness of combining TCN and GRU for robust classification.

Overall, the stacking ensemble and GRU models showed superior performance, with the TCN-GRU hybrid offering competitive results and underscoring the potential of ensemble and hybrid methods for accurate metro train failure classification.

The performance of ensemble machine learning models, deep learning models, and a hybrid model for metro train Remaining useful life (RUL) prediction was evaluated using RMSE, R2-score, MSE, and MAE metrics, as summarized in Table 2.

TABLE II. REMAINING USEFUL LIFE (RUL) REGRESSION MODELS

Models	RMSE	R2-score	MSE	MAE
Random Forest	0.42	0.81	0.18	0.29
XGBoost	0.43	0.80	0.19	0.32
Bagging	0.42	0.81	0.18	0.29
Stacking	0.42	0.81	0.18	0.30
CNN	0.51	0.73	0.27	0.37
LSTM	0.28	0.91	0.08	0.19
GRU	0.28	0.91	0.08	0.18
TCN	0.30	0.91	0.09	0.20
TCN-GRU	0.26	0.92	0.07	0.19

Our regression models demonstrated superior performance compared to existing studies for Remaining Useful Life (RUL) prediction in metro trains. However, the results may vary when applied to other machines. The proposed models addressed key research objectives by introducing the hybrid TCN-GRU model, effectively reducing prediction error while minimizing training time and mitigating overfitting. Notably, the TCN model required less training time compared to CNN, and the LSTM model was faster than GRU, while both TCN and GRU effectively handled overfitting.

Among the ensemble models, Random Forest, Bagging, and Stacking achieved similar performance with RMSE of 0.42, R^2-score of 0.81, MSE of 0.18, and MAE between 0.29 and 0.30. XGBoost slightly underperformed with RMSE of 0.43 and MSE of 0.19.

In deep learning, LSTM and GRU outperformed the ensemble methods with RMSE of 0.28, R^2-score of 0.91, and MSE of 0.08. GRU achieved the lowest MAE of 0.18, demonstrating its superior accuracy. The TCN model also performed well, with RMSE of 0.30, R^2-score of 0.91, and MAE of 0.20.

The hybrid TCN-GRU model achieved the best overall performance, with RMSE of 0.26, R^2-score of 0.92, MSE of 0.07, and MAE of 0.19. This highlights the combined strengths of TCN and GRU, resulting in improved accuracy and robustness.

Overall, the GRU and hybrid TCN-GRU models outperformed traditional ensemble methods, demonstrating the potential of advanced deep-learning techniques for accurate RUL prediction in metro trains.

B. XAI Models Interpretation and Explainability

In our study, we did not apply XAI methods like LIME and SHAP to deep learning and hybrid models due to the challenges posed by time series data. Unlike tabular or image data, time series data is sequential, where each time step influences subsequent ones. Methods like LIME, which perturb individual features, can disrupt the sequence, leading to misleading explanations. Similarly, SHAP struggles to capture the complex temporal dependencies between past and future observations.

However, we applied LIME and SHAP to the best-performing ensemble machine learning model, XGBoost, for both failure classification and RUL prediction due to its lower training time compared to other models. While LIME provides local interpretations, it lacks overall feature contribution insights. In contrast, SHAP offers global interpretability by providing the general contribution of each feature. Therefore, we presented the SHAP results to highlight the key features influencing model predictions in both classification and regression tasks.

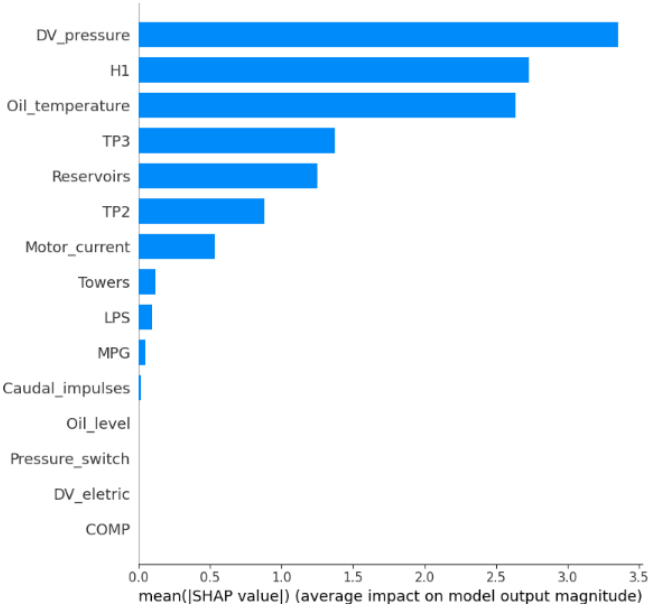


Fig. 4. SHAP Global Interpretation for Ensemble Classification Model

According to Figure 4, parameters such as DV_pressure, H1, Oil_temperature, TP3, Reservoirs, TP2, Motor_current, etc. show the highest mean absolute SHAP values. This signifies that these parameters are the most influential in predicting failures (class 1) within the metro train air production unit.

As shown in Figure 5, parameters such as Oil_temperature, Motor_current, Reservoirs, DV_pressure, Caudal_impulses, H1, TP2, etc. demonstrate the highest mean absolute SHAP values. This signifies that these elements lead to a reduction in the remaining useful life (RUL) of the metro train air production unit, indicating an accelerated decline in the machine's health and a heightened risk of impending failure.

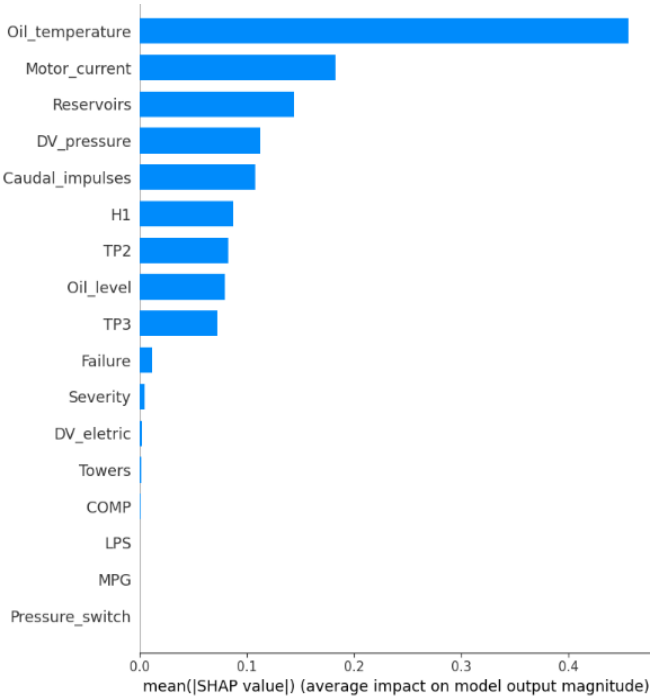


Fig. 5. SHAP Global Interpretation for Ensemble Regression Model

Adjusting the parameters recognized in Figures 4 and 5 enables maintenance teams to emphasize preventive measures, thereby potentially decreasing the frequency of failures and prolonging the lifespan, i.e. remaining useful life (RUL) of metro trains.

C. *Limitations of our study*

The methodology proposed in this study is limited in scope, as it has been applied solely to the metro train air production unit, excluding other factors that could influence metro train failure prediction and not utilizing data from other machine sensors. Real-world data collection faces challenges, such as incomplete, inconsistent, or noisy data, as well as difficulties in obtaining proprietary or sensitive operational data, complicating the preprocessing of datasets. Furthermore, the methodology has not been validated with real-time sensor data, leaving its practical applicability uncertain. Some failure prediction models require significant training time due to the complexity of the algorithms and the size of the datasets, potentially limiting scalability and real-time implementation. Additionally, the interpretability of deep learning models is constrained, as the study does not incorporate XAI techniques to understand the time-series patterns learned by models like LSTM, GRU, etc., due to the reasons outlined in our research.

V. Conclusion

This study highlights the effectiveness of ensemble machine learning models and hybrid deep learning algorithms in improving failure classification and remaining useful life estimation for metro trains. By addressing key limitations of prior approaches—such as imbalanced datasets, handling large time-series data, overfitting, and reliance on synthetic data—we employed resampling techniques and evaluated various models to identify optimal solutions. Our proposed ensemble and hybrid models, including stacking and TCN-GRU, demonstrate superior accuracy and reduced error, supported by explainable AI tools like LIME and SHAP, which identify critical failure factors. Ensemble methods like Stacking demonstrated robust performance for failure classification, achieving up to 99.95% accuracy. Meanwhile, hybrid TCN-GRU models achieved the best results in remaining useful life (RUL) prediction with an RMSE of 0.26 and an R^2-score of 0.92. The practical implications for the metro rail industry include reduced maintenance costs, enhanced reliability, optimized performance, and safer operations.

Future work should focus on expanding datasets for other machines for improved generalizability, more effective resampling techniques, reducing model training times facilitating real-time deployment and faster decision-making, and developing adaptive learning approaches that can adjust to dynamic and real-time applications. Additionally, integrating user-friendly interfaces with real-time health monitoring and failure notifications can further enable proactive maintenance and informed decision-making.

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